

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T25 · Solving Quadratics

With Answers

Questions with full worked solutions

■ 19 pages

FAMILY MAP

Solving Quadratic Equations

F01

Solve a quadratic by factorisation

9 Q | 23 marks

Question 22

4MA1-2H Q11

ORIGINAL QUESTION

3 marks

11 (i) Factorise $x^2 + 9x - 22$

.....
(2)

(ii) Hence, solve $x^2 + 9x - 22 = 0$

.....
(1)

Worked Solution 22

Factorise, hence solve

MODULAR UNIT 1

3 marks

Factorise

$$x^2 + 9x - 22 = (x - 2)(x + 11).$$

Roots

$$(x - 2)(x + 11) = 0.$$

Final answer

$$(x - 2)(x + 11); \quad x = 2, -11$$

Question 05

4MA1-1H Q6

ORIGINAL QUESTION

3 marks

(d) (i) Factorise $x^2 + 9x - 22$

.....
(2)

(ii) Hence solve $x^2 + 9x - 22 = 0$

.....
(1)

Worked Solution 05

Factorise then solve

MODULAR UNIT 1

3 marks

Factorise

$$x^2 + 9x - 22 = (x + 11)(x - 2).$$

Use the zero-product rule

$$x + 11 = 0 \text{ or } x - 2 = 0.$$

Final answer

$$x = -11 \text{ or } x = 2$$

Original question 07

4MA1-1H Q8 M01, M02

MODULAR UNIT 1

3 marks

8 (a) (i) Factorise $x^2 + 5x - 24$
(2)(ii) Hence, solve $x^2 + 5x - 24 = 0$
(1)

Worked solution 07

Factorise a quadratic / Solve the factorised equation

MODULAR UNIT 1

3 marks

M01 – Factorise a quadratic**Find two numbers**

The product is -24 and the sum is 5, so use 8 and -3.

Write the factors

$$x^2 + 5x - 24 = (x + 8)(x - 3).$$

Final answer

$$(x+8)(x-3)$$

M02 – Solve the factorised equation**Set each factor to zero**

$$x + 8 = 0 \text{ or } x - 3 = 0.$$

State both roots

$$x = -8 \text{ or } x = 3.$$

Final answer

$$x = -8 \text{ or } x = 3$$

Original question 21

4MA1-2H Q11 Q11(i), Q11(ii)

ORIGINAL QUESTION**3 marks****11** (i) Factorise $x^2 + 5x - 24$

(2)(ii) Hence, solve $x^2 + 5x - 24 = 0$

(1)

Worked solution

Family: Factorise a monic quadratic • Variant: Negative constant monic quadratic

METHOD / WORKING

MODULAR UNIT 1 • JUNE 2022 DETAILED SOLUTIONS
Worked solution 21 MODULAR UNIT 1
Chapter: Factorising; Solving Quadratic Equations 3 marks
Q11(i) – Chapter: Factorising
Family: Factorise a monic quadratic • Variant: Negative constant monic quadratic
Method
The required numbers have product -24 and sum 5, so they are 8 and -3.
Why: Match the constant term and middle coefficient.
Method
<i>Therefore</i> $x^2 + 5x - 24 = (x - 3)(x + 8)$.
Why: Write the monic quadratic as two linear factors.
Final answer
factorisation: $(x - 3)(x + 8)$
Q11(ii) – Chapter: Solving Quadratic Equations
Family: Solve a quadratic by factorisation • Variant: Monic quadratic roots by factors
Method
From $(x - 3)(x + 8) = 0$, $x = 3$ or $x = -8$.
Why: Use the zero-product rule on the factors from part (i).
Final answer
roots: $x = 3$ or $x = -8$

FINAL ANSWER

Original question 05

4MA1-1H Q6 Q6(b)(i), Q6(b)(ii)

ORIGINAL QUESTION**3 marks**(b) (i) Factorise $y^2 - 2y - 35$

(2)(ii) Hence, solve $y^2 - 2y - 35 = 0$

(1)

Worked solution

Family: Factorise a non-monic or special quadratic • Variant: Factorise as the first part of a factor-then-solve

METHOD / WORKING

MODULAR UNIT 1 • JANUARY 2022 DETAILED SOLUTIONS
Worked solution 05 MODULAR UNIT 1
Chapter: Factorising; Solving Quadratic Equations 3 marks
Q6(b)(i) – Chapter: Factorising
Family: Factorise a non-monic or special quadratic • Variant: Factorise as the first part of a factor-then-solve
chain
Method
We need two numbers with product -35 and sum -2 : -7 and $+5$.
Why: The constant term is -35 and the x -coefficient is -2 .
Method
$y^2 - 2y - 35 = (y - 7)(y + 5)$.
Why: Expand the brackets to check.
Final answer
factorisation: $(y-7)(y+5)$
Q6(b)(ii) – Chapter: Solving Quadratic Equations
Family: Solve a quadratic by factorisation • Variant: Monic quadratic roots by factors
Method
$(y-7)(y+5)=0$, so $y=7$ or $y=-5$.
Why: Use the zero-product rule; this follows the factorisation in part (i).
Final answer
roots: $y = 7$ or $y = -5$

FINAL ANSWER

- (b) Solve $x^2 - 3x - 40 = 0$
Show clear algebraic working.

(3)

Worked solution

January 2020 | Question 7 | 3 marks

Family: Solve a quadratic by factorisation | Variant: Monic quadratic roots by factors

METHOD / WORKING

1. $x^2 - 3x - 40 = (x - 8)(x + 5)$.

2. Set each factor to zero.

3. Roots are $x=8$ and $x=-5$.

FINAL ANSWER

 $x=8$ or $x=-5$

(b) (i) Factorise $x^2 + 5x - 36$

(2)

(ii) Hence, solve $x^2 + 5x - 36 = 0$

(1)

Worked solution

January 2021 | Question 9 | 1 marks

Family: Solve a quadratic by factorisation • Variant: Monic quadratic roots by factors

Method / working

1. Set each factor to zero: $x+9=0$ or $x-4=0$.

Final answer: $x=-9$ or $x=4$

9 (a) Factorise fully $25a^4c^7d + 45a^9c^3h$

(b) Solve $(2x + 5)^2 = (2x + 3)(2x - 1)$

.....
(2)

$x =$
(3)

Worked solution

June 2021 | Question 9 | 3 marks

Family: Solve a quadratic by factorisation | Variant: Quadratic rearranged before factorising

METHOD / WORKING

1. *Expand both sides*: $4x^2 + 20x + 25 = 4x^2 + 4x - 3$.

2. Collect x terms and constants: $16x = -28$.

3. Hence $x = -28/16 = -7/4 = -1.75$.

FINAL ANSWER

$$x = -7/4$$

9 (i) Factorise $x^2 + 2x - 24$

(2)

(ii) Hence solve $x^2 + 2x - 24 = 0$

(1)

Worked solution

June 2021 | Question 9 | 1 marks

Family: Solve a quadratic by factorisation • Variant: Monic quadratic roots by factors

Method / working

1. Set each factor to zero: $x+6=0$ or $x-4=0$.

Final answer: $x=-6$ or 4